

Solution Requirements

Date	29 June 2026
Team ID	
Project Name	Rising Waters
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR)** define specific behaviours, functions, or features of the system (What the system should *do*).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviours (How the system should *be*).

Fill out the descriptions and necessary details for each category provided below

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Authentication	Users securely log in using email and password. Admin authentication is required to manage the system	High

2	Authorization levels	Admins can manage datasets, users, and predictions. Users can view flood predictions, weather information, and alerts only	High
4	Transactions processing	Collects weather data, preprocesses it, runs the ML model, and generates flood risk predictions in real time	High
5	Reporting	Displays flood prediction reports, risk levels, rainfall trends, and downloadable prediction summaries	Medium
6	Business rules, etc.	Flood alerts are generated when rainfall and weather conditions exceed predefined thresholds. The AI model classifies risk as Low, Medium or High.	High
7	Compliance to laws or regulations	User information is protected. Weather data is used according to public data policies and privacy guidelines	Medium

8	Other	Sends Emergency notification and provide recommendations	Medium
---	-------	--	--------

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
------	--------------	-------------------------	-------------------------------------

1	Performance & Speed	The system should generate flood predictions quickly.	Prediction within 5 seconds
2	Scalability	Support multiple users and increase weather datasets without performance issues.	Supports 1000+ concurrent users
3	Security & Data Privacy	secure user accounts and protect stored weather and prediction data.	Password encryption and secure database
4	Reliability & Availability	The system should be available during emergencies with minimal downtime.	99% uptime

5	Usability & Accessibility	User-friendly interface with clear flood warnings, maps, and dashboards.	Usability & Accessibility Easy navigation and mobile-friendly
6	Other	Other Easy maintenance and future integration with IoT sensors and additional weather APIs.	Modular and extensible architecture